

EES 201: Introduction to Earth Sciences

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Source: Department of Earth and Planetary Sciences –
Northwestern - Northwestern University

■ What is sedimentology

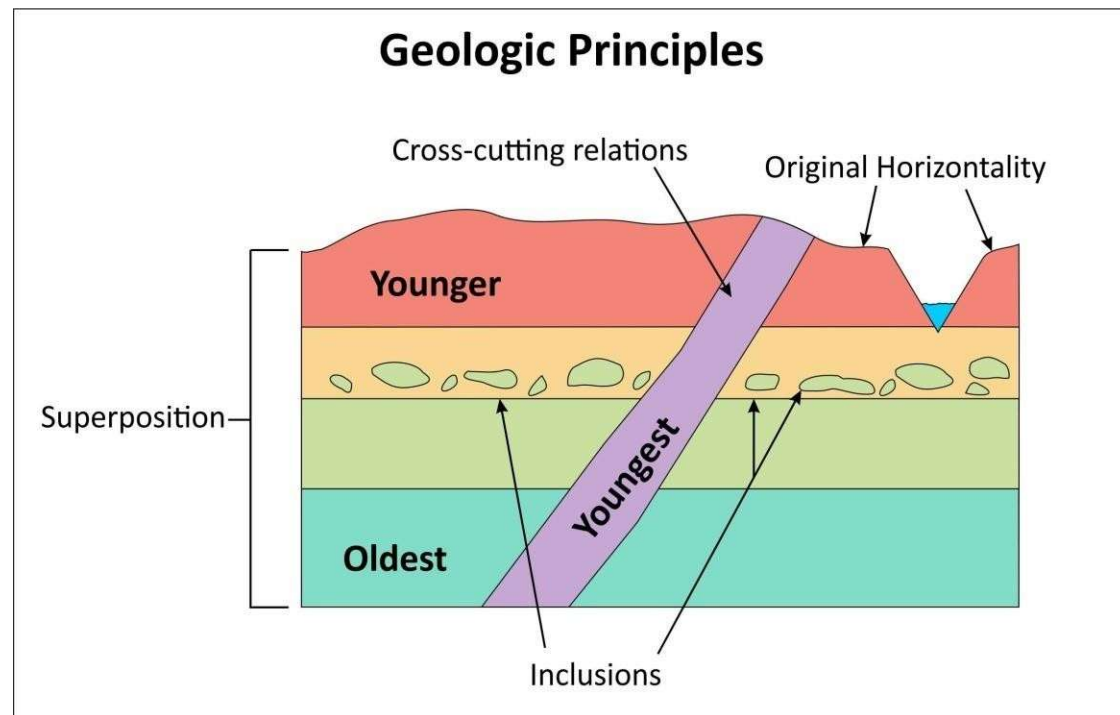
- Study of the processes of **formation**, **transport** and **deposition** of material that accumulates as sediment in continental and marine environments and eventually forms **sedimentary rocks** (G. Nichols).

Brief History

- The history of Sedimentology dated back to about 1500 AD.

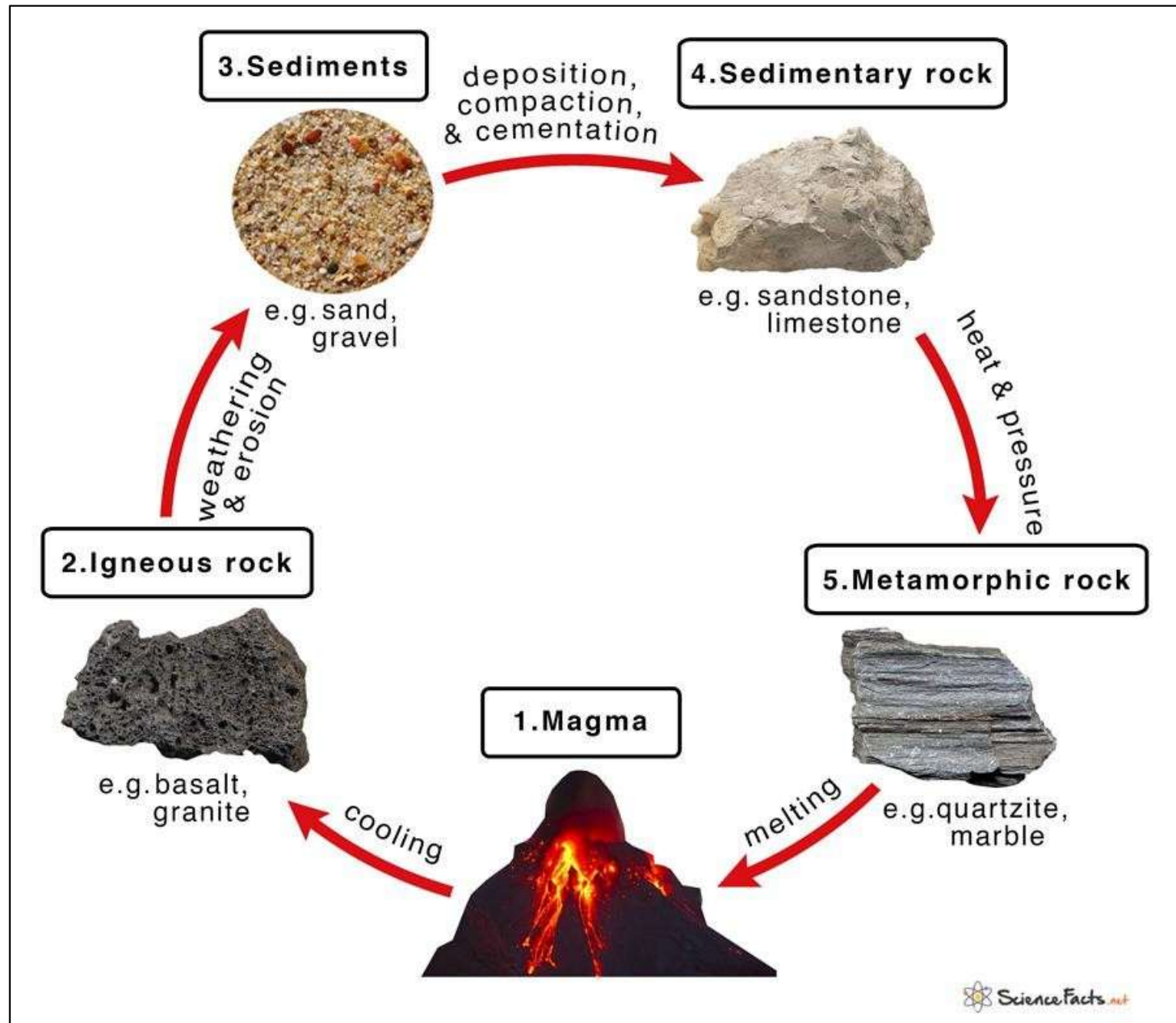
- Principle of superposition (Nicolas Steno, 1669).

- *“within a sequence of layers of sedimentary rock, the oldest layer is at the base and that the layers are progressively younger with ascending order in the sequence”*



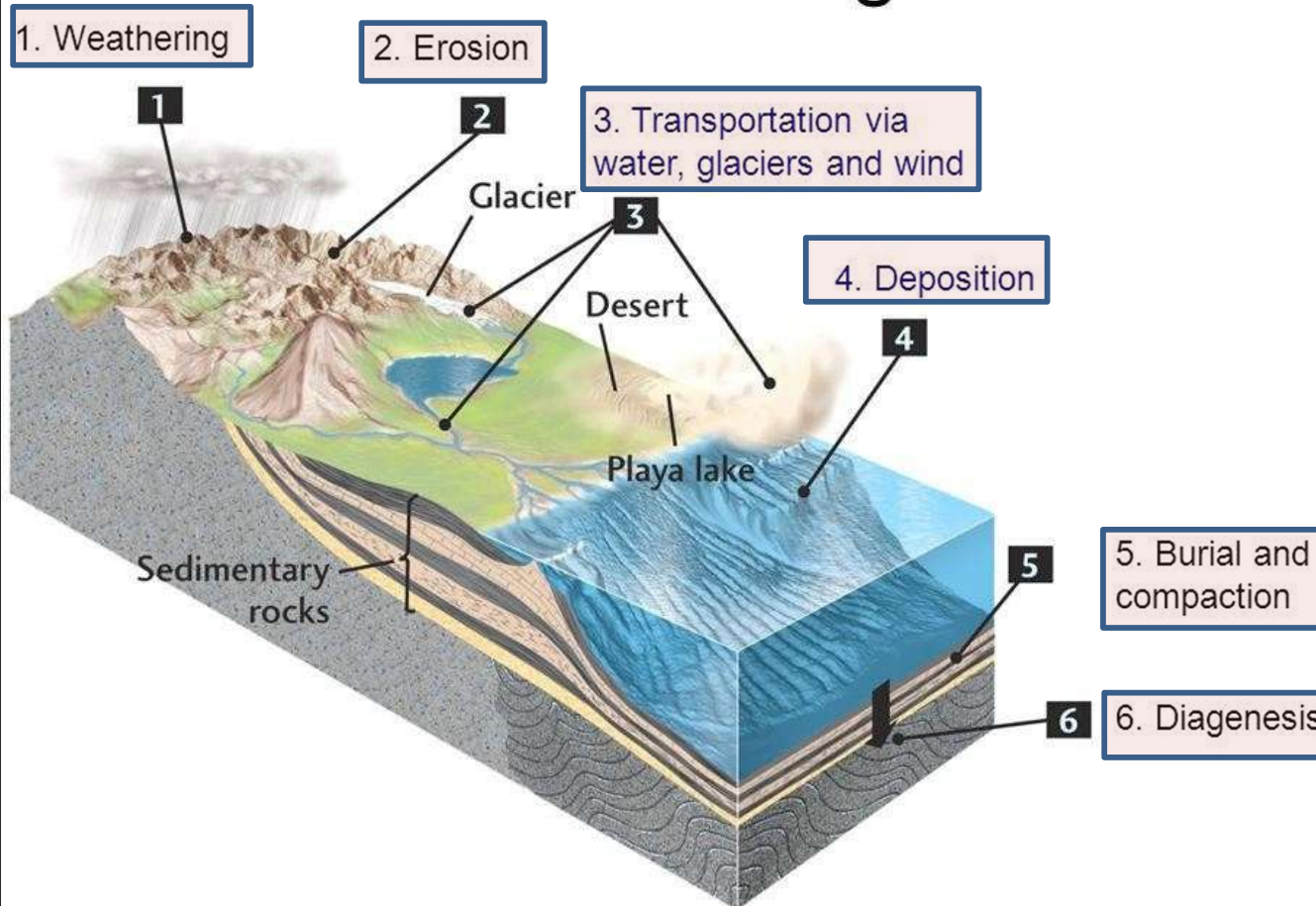
- Uniformitarianism (“the present is the key to the past”) (Charles Lyell, 1830): is the assumption that the same natural laws and processes that operate in our present-day scientific observations have always operated in the universe in the past and apply everywhere in the universe

Sedimentary rock



Sedimentary rock

Sediment stages



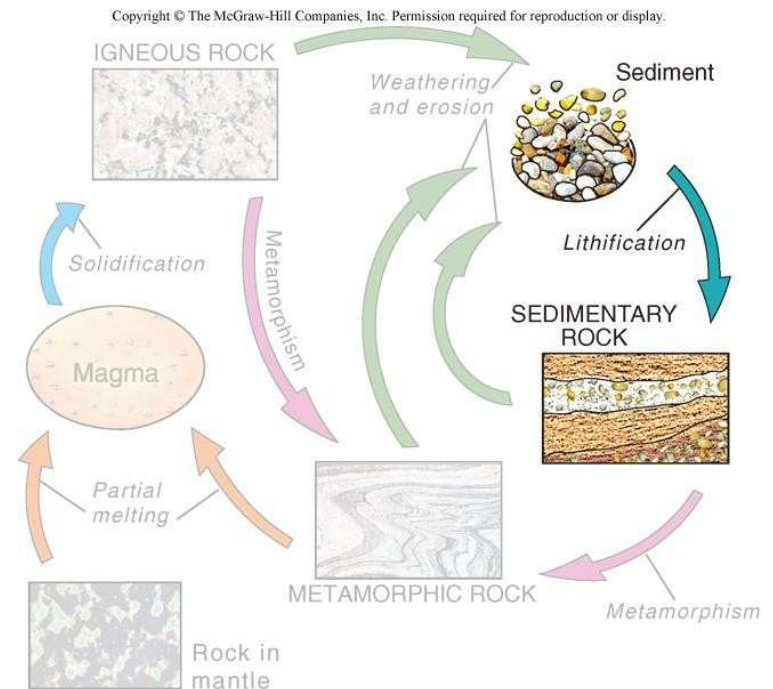
- Weathering & Erosion
 - Decomposition
 - Disintegration
- Transportation and deposition
- Diagenesis and lithification
- Compaction and cementation.

Introduction to Sedimentary rock

- *Sediment* - loose, solid particles originating from:
 - Weathering and erosion of pre-existing rocks
 - Chemical precipitation from solution, including secretion by organisms in water
- Classified by *particle size*

- Boulder - >256 mm
- Cobble - 64 to 256 mm
- Pebble - 2 to 64 mm
- Sand - 1/16 to 2 mm
- Silt - 1/256 to 1/16 mm
- Clay - <1/256 mm

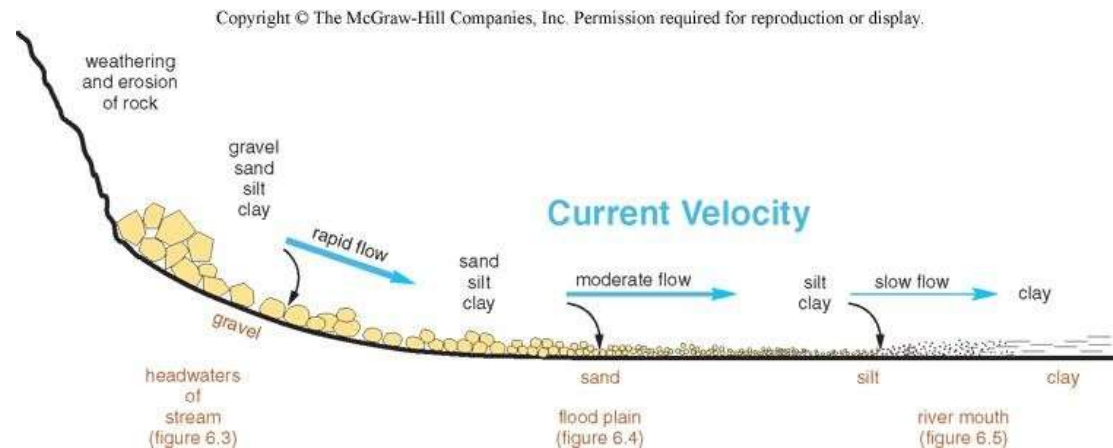
Gravel



Introduction to Sedimentary rock

- *Transportation*

- Movement of sediment away from its source, typically by water, wind, or ice
- *Rounding* of particles occurs due to abrasion during transport
- *Sorting* occurs as sediment is separated according to grain size by transport agents, especially running water
- Sediment size decreases with increased transport distance



Introduction to Sedimentary rock

- *Deposition*

- Settling and coming to rest of transported material
- Accumulation of chemical or organic sediments, typically in water
- *Environment of deposition* is the location in which deposition occurs
 - Deep sea floor
 - Beach
 - Desert dunes
 - River channel
 - Lake bottom



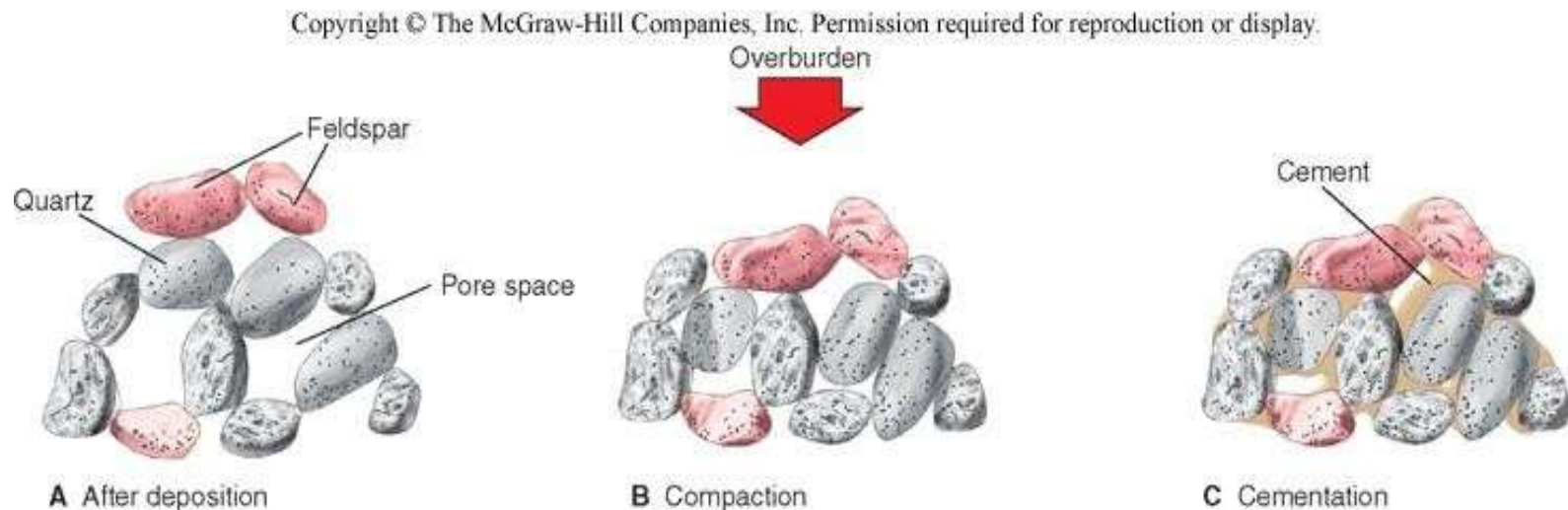
Introduction to Sedimentary rock

- *Preservation*

- Sediment must be preserved, as by burial with additional sediments, in order to become a sedimentary rock

- *Lithification*

- General term for processes converting loose sediment into sedimentary rock
- Combination of *compaction* and *cementation*



Classification of Sedimentary rock

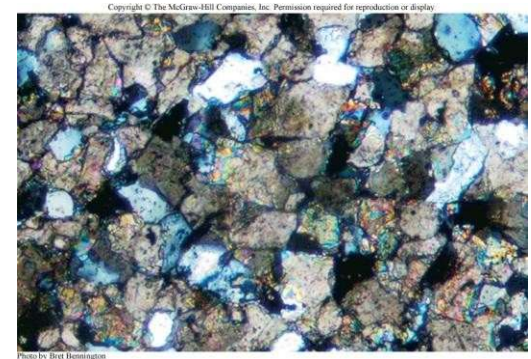
- *Detrital (clastic) sedimentary rocks*

- Most common sedimentary rock type
- Form from cemented sediment grains that come from pre-existing rocks



- *Chemical sedimentary rocks*

- Have crystalline textures
- Form by precipitation of minerals from solution



- *Organic sedimentary rocks*

- Accumulate from remains of organisms



Classification of Sedimentary rock: Clastic rocks

- *Breccia and Conglomerate*

- *Coarse-grained clastic* sedimentary rocks



- *Sandstone*

- *Medium-grained clastic* sedimentary rock
- Types determined by composition
 - *Quartz sandstone* - >90% quartz grains
 - *Arkose* - mostly feldspar and quartz grains
 - *Graywacke* - sand grains surrounded by dark, fine-grained matrix, often clay-rich



Classification of Sedimentary rock: Clastic rocks

- *Shale*

- Fine-grained clastic sedimentary rock
- Splits into thin layers (*fissile*)
- Silt- and clay-sized grains



- *Siltstone*

- Slightly coarser-grained than shales
- Lacks fissility

- *Claystone*

- Predominantly clay-sized grains; non-Fissile

- *Mudstone*

- Silt- and clay-sized grains; massive/blocky



Classification of Sedimentary rock: Chemical rocks

- *Carbonates*

- Contain CO_3 as part of their chemical composition
- *Limestone* is composed mainly of *calcite*
 - Most are *biochemical*, but can be *inorganic*
 - Often contain easily recognizable fossils
 - Chemical alteration of limestone in Mg-rich water solutions can produce *dolomite*



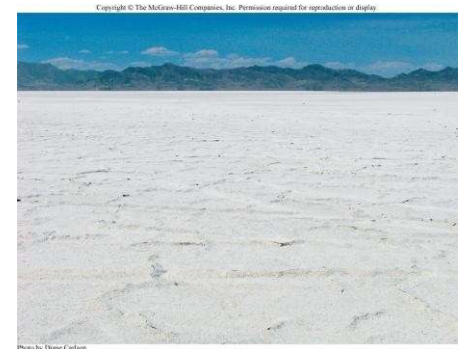
- *Chert*

- Hard, compact, fine-grained, formed almost entirely of silica
- Can occur as layers or as lumpy nodules within other sedimentary rocks, especially limestones



- *Evaporites*

- Form from evaporating saline waters (lake, ocean)
- Common examples are rock gypsum, rock salt



Classification of Sedimentary rock: Organic rocks

- Coal
 - Sedimentary rock forming from compaction of partially decayed plant material
 - Organic material deposited in water with low oxygen content (i.e., stagnant)
- Oil and natural gas
 - Originate from organic matter in marine sediment
 - Subsurface “cooking” can change organic solids to oil and natural gas
 - Can accumulate in porous overlying rocks

Sedimentary structures

- Features within sedimentary rocks produced during or just after sediment deposition
- Provide clues to how and where deposition of sediments occurred
- *Bedding*
 - Series of visible layers within a rock
 - Most common sedimentary structure
- *Cross-bedding*
 - Series of thin, inclined layers within a horizontal bed of rock
 - Common in sandstones
 - Indicative of deposition in ripples, bars, dunes, deltas



Sedimentary structures

- *Ripple marks*

- Small ridges formed on surface of sediment layer by moving wind or water



- *Graded bedding*

- Progressive change in grain size from bottom to top of a bed



- *Mud cracks*

- Polygonal cracks formed in drying mud



- *Fossils*

- Traces of plants or animals preserved in rock
- Hard parts (shells, bones) more easily preserved as fossils

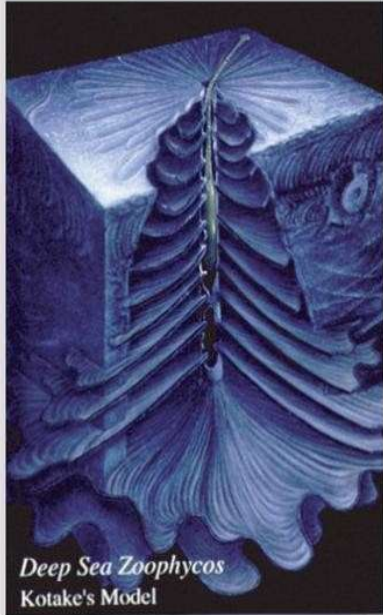


Trace Fossils

- **Domichnia**, dwelling structures reflecting the life position of the organism that created it.
- **Fodinichnia**, three-dimensional structures left by animals which eat their way through sediment, such as deposit feeders;
- **Pascichnia**, feeding traces left by grazers on the surface of a soft sediment or a mineral substrate;
- **Cubichnia**, resting traces, in the form of an impression left by an organism on a soft sediment;
- **Repichnia**, surface traces of creeping and crawling.

ZOOPHYCOS ICHNOFACIES

Characteristic Forms of Zoophycos



NEREITES ICHNOFACIES

Characteristic Ichnogenera



Paleodictyon



Deep Sea Cores:
Zoophycos, Chondrites,
and *Planolites*



Nereites



Cosmorhaphe



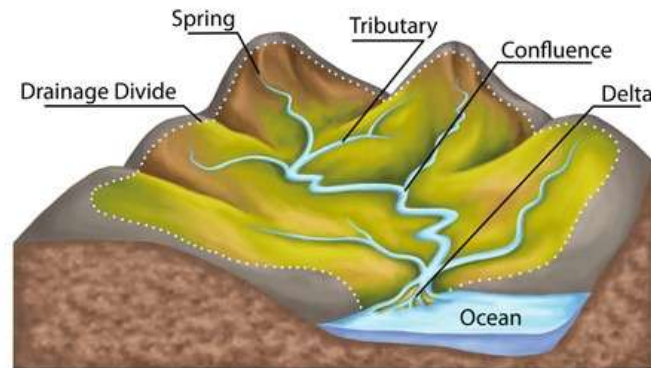
Helminthoida

Outline

- Fluvial systems: Introduction
 - *Grain size : basics*
- Fluvial processes
 - *Erosion*
 - *Transportation*
 - *Deposition*
- River forms
 - *Braided river*
 - *Meandering river*
 - *Flood plains*
 - *Terraces*
- Case study
 - *NW Himalaya*
 - *Tso Moriri*

Fluvial system: Consists of a network of channels and associated environments that transport sediments from a **drainage basin** to a **depositional basin**

Drainage basin also known as watersheds or catchments.

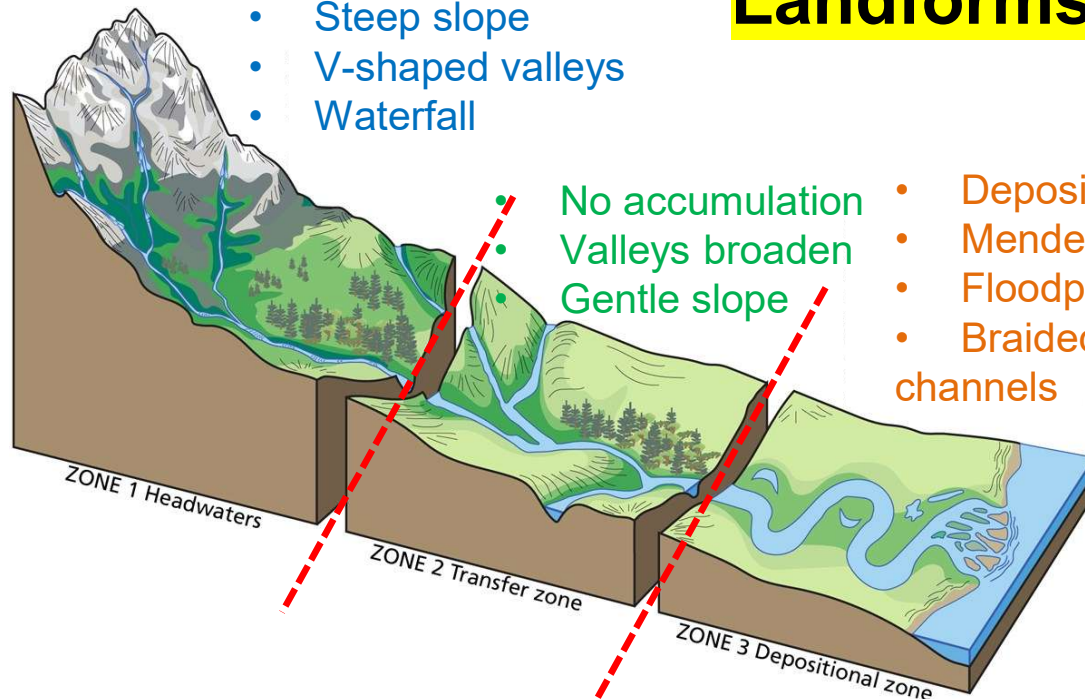


Depositional basin: sediment and water get accumulated



Landforms

- Erosion
- Steep slope
- V-shaped valleys
- Waterfall

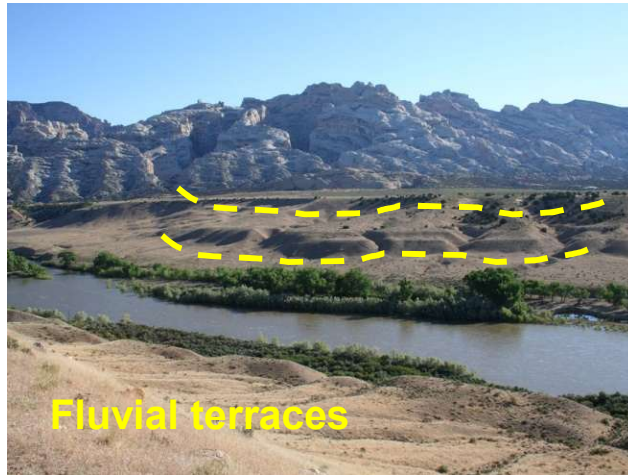


Fluvial system: Consists of a network of channels and associated environments that transport sediments from a drainage basin to a depositional basin

Landforms



V-shaped valley



Fluvial terraces



Alluvial fans



Meandering river



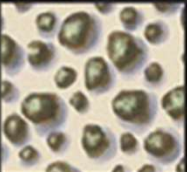
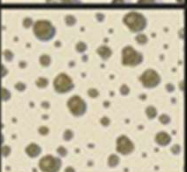
Braided river



Flood

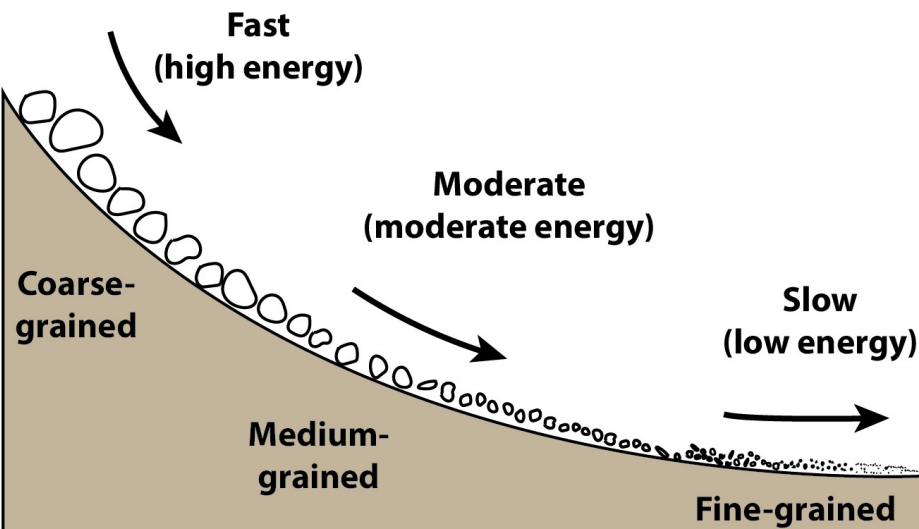
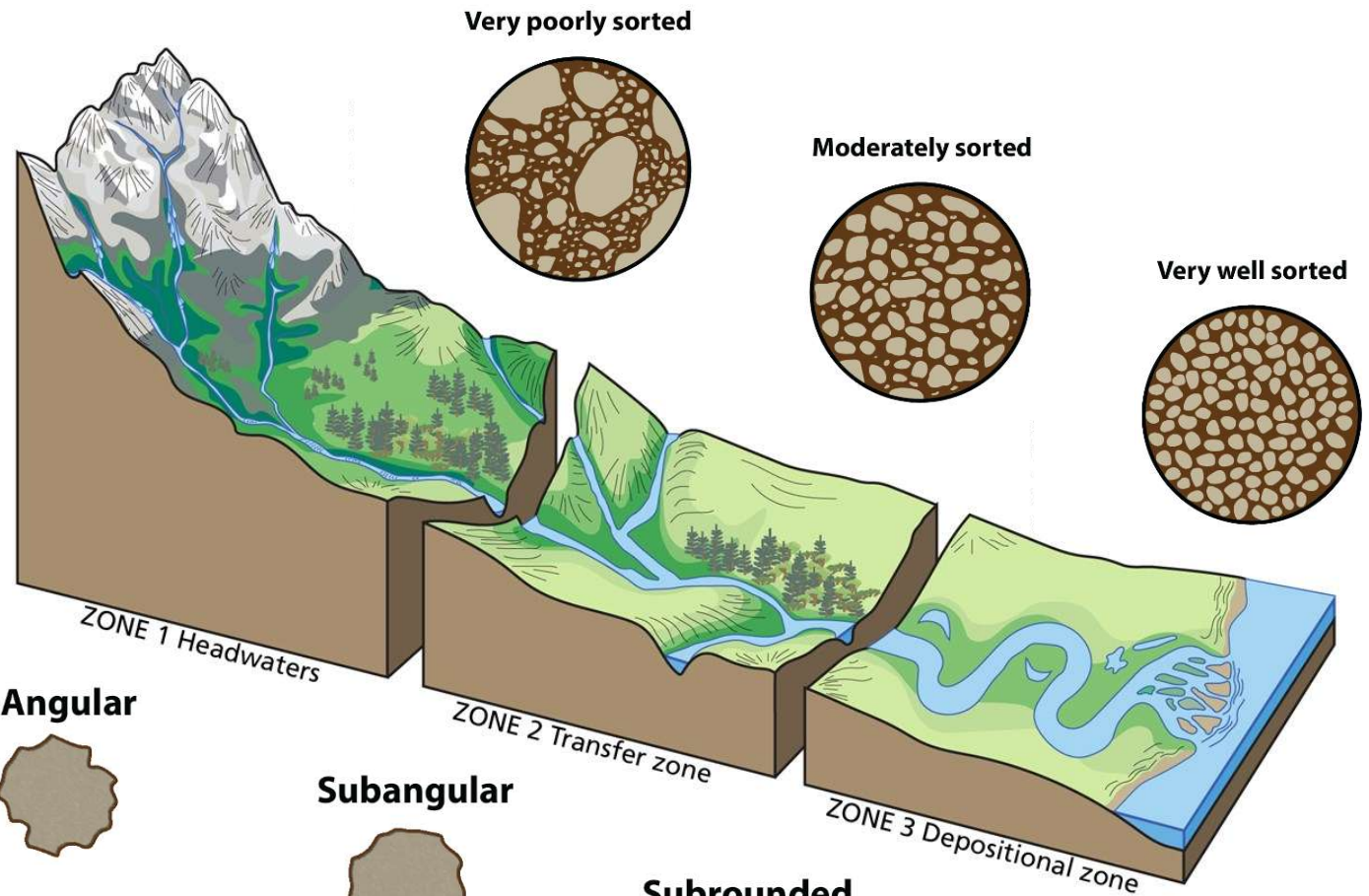
Grain size

- Significant changes in landscape and sediment characteristics
- Krumbein (1934)
- $\phi = -\log_2 d$ (grain diameter in mm) naturally occurring groups
 - **Gravel** ~ rock fragments
 - **Sand** ~ individual mineral grains (particulate residues)
 - **Mud** ~ particulate residues +/- chemical weathering products
 - **Clay** ~ chemical weathering products (clay minerals, etc.)

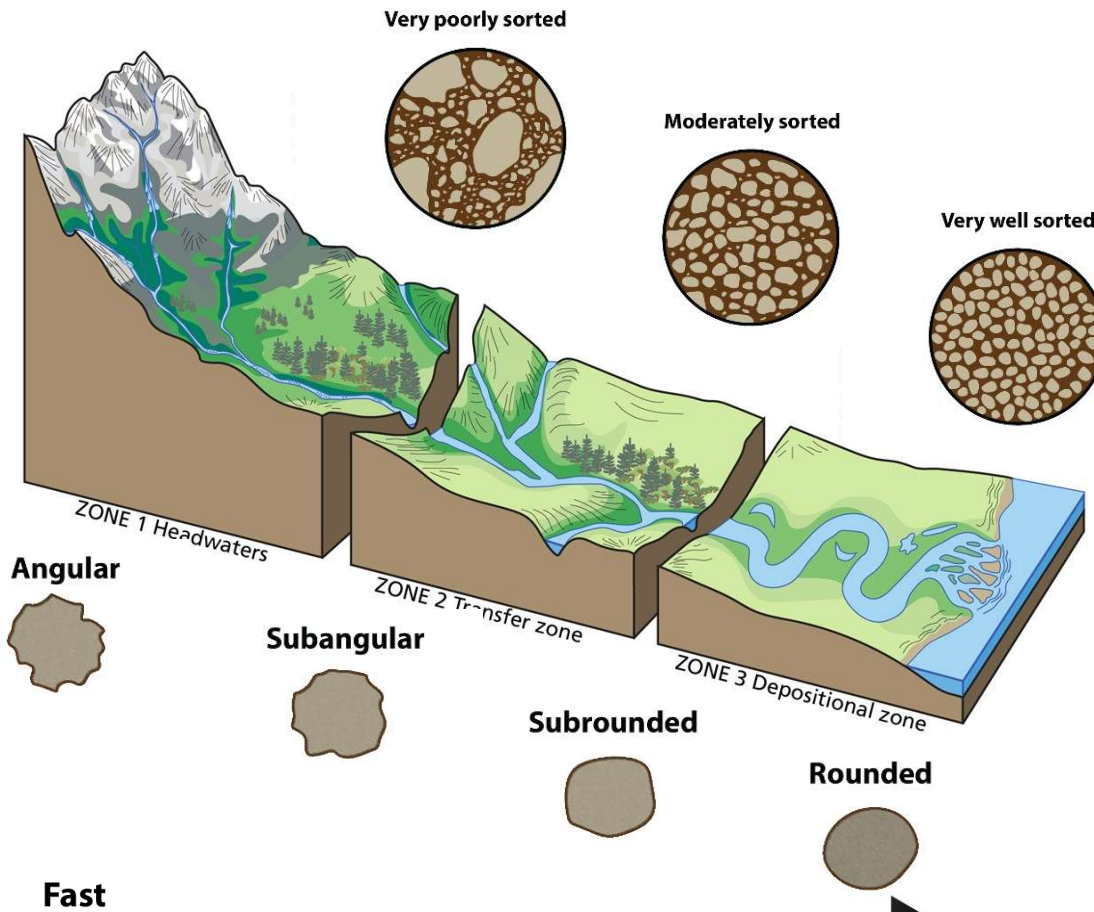
Wentworth Size Class		mm scale	phi scale
Boulder > 256 mm (-8 to -12 ϕ)	Pebbles	256 to 4	 -8 to -2
	Gravel	4 to 2	 -2 to -1
	Very coarse and Coarse sand	2 to 0.5	 -1 to 1
Fine and Very fine sand	Medium sand	0.5 to 0.25	 1 to 2
	Very fine sand	0.25 to 0.06	 2 to 4
	Silt	0.06 to 0.004	 4 to 8
	Clay	< 0.004	 > 8.00

Fluvial system

Sediment



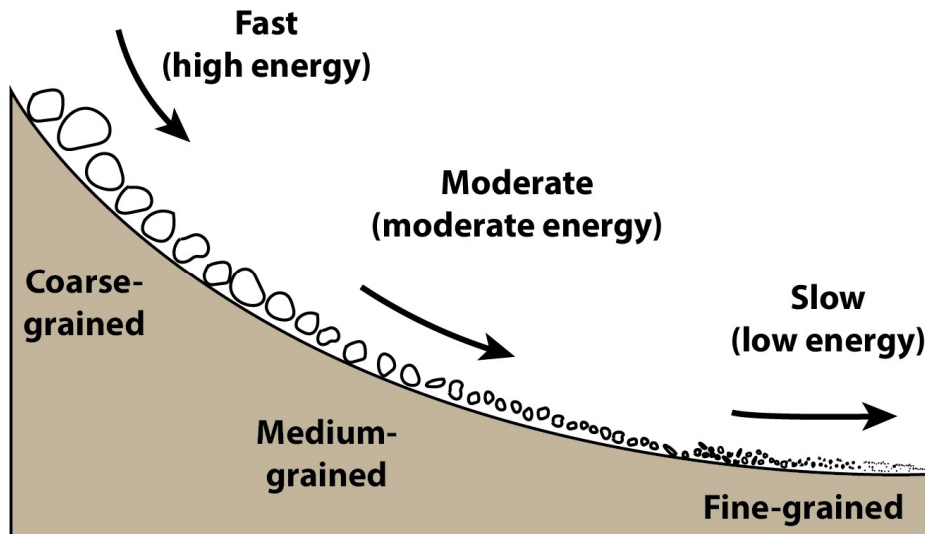
Downstream changes in alluvium grains



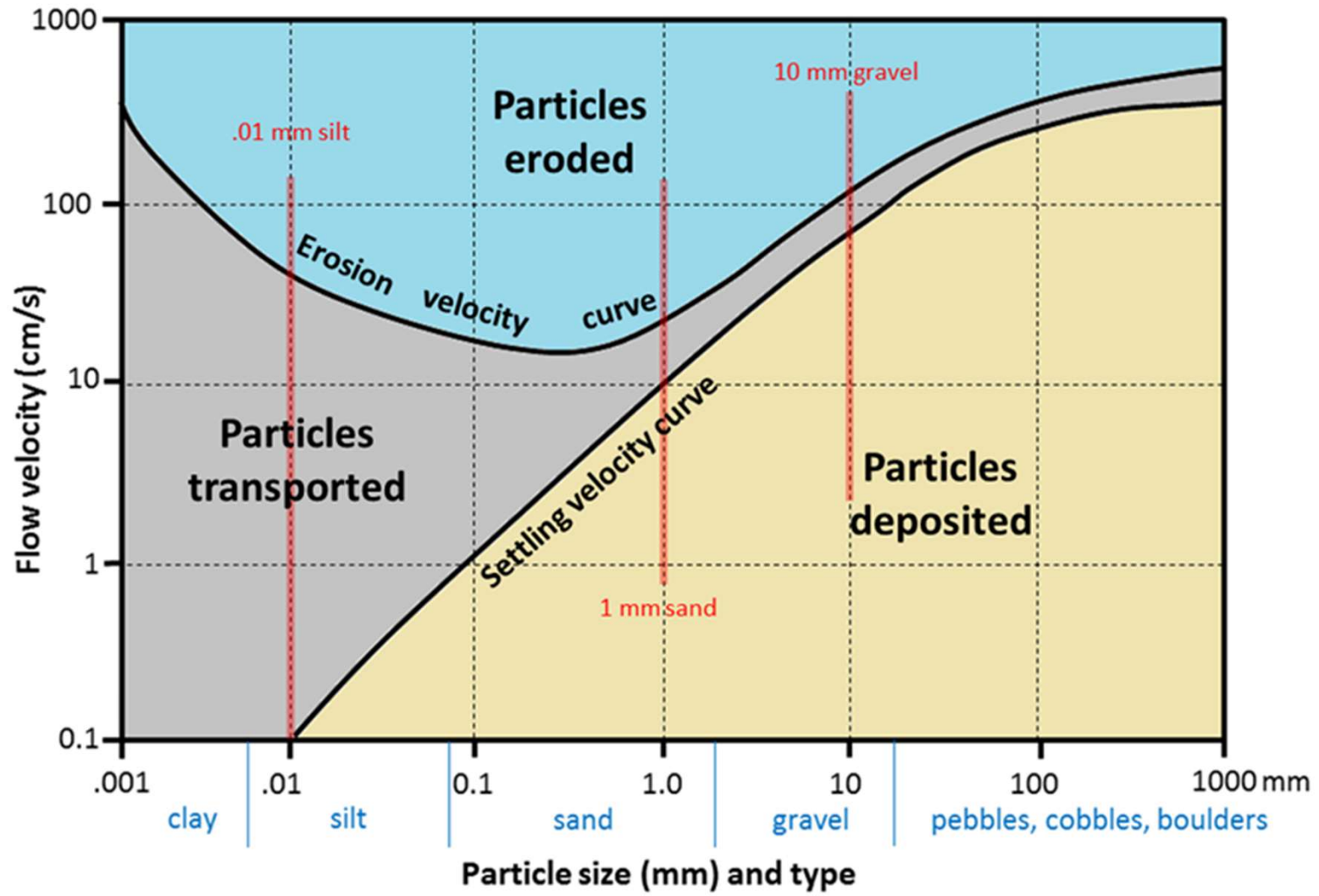
Particles are **large** and **irregular**, and consist of a variety of minerals, including the least resistant.

Particles are **mid-sized** and of **intermediate roundness**, and include resistant and nonresistant minerals.

Particles are **small** and nearly **spherical**, and consist mainly of the most resistant minerals, such as quartz.

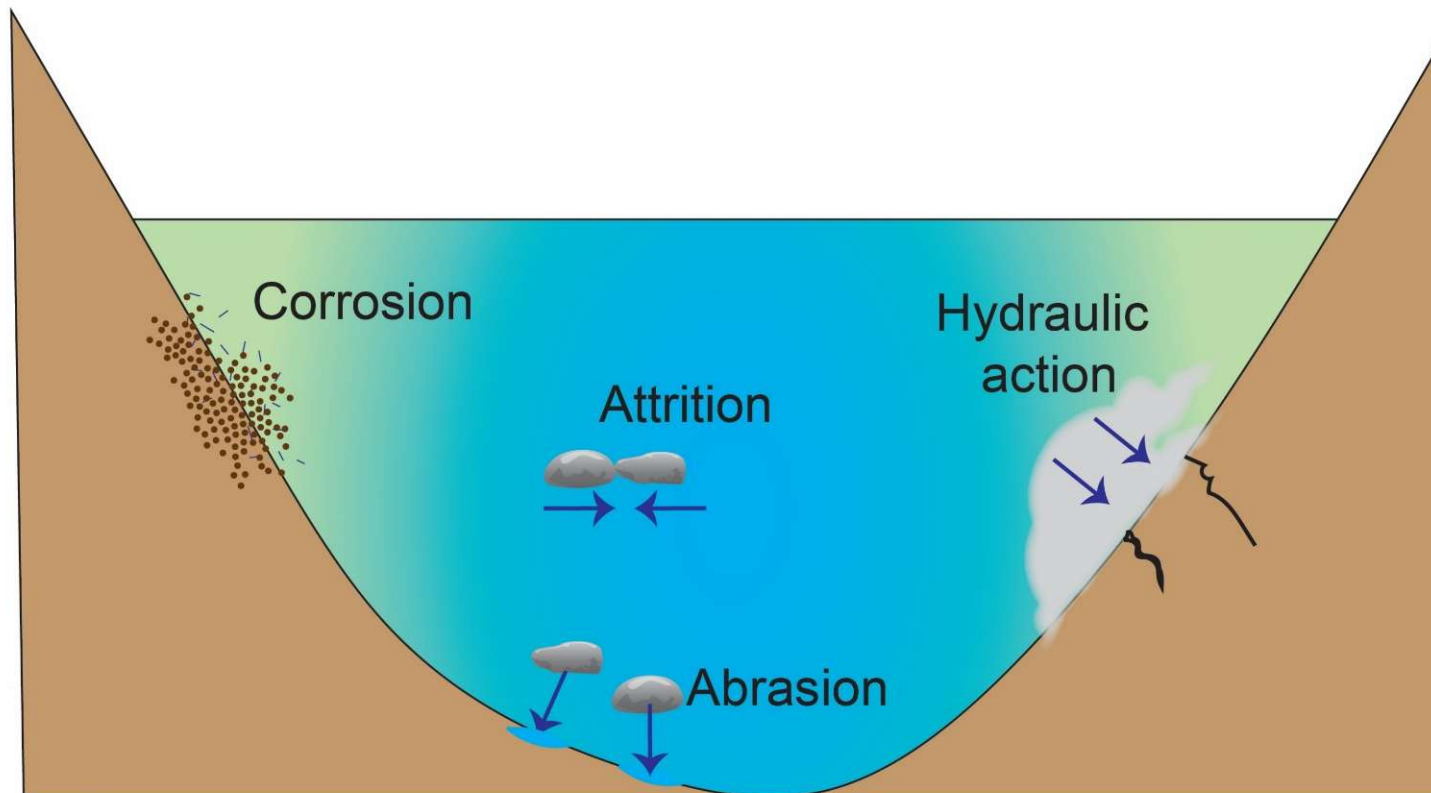


Fluvial processes



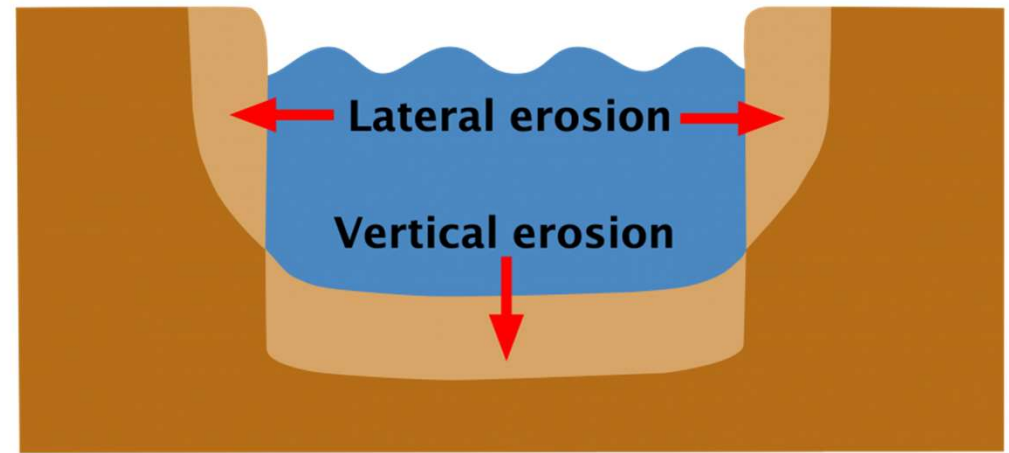
The Hjulström-diagram showing the relationships between particle size and the tendency to be eroded, transported, or deposited at different current velocities

- **Fluvial erosion** is the detachment of material of the river bed and the sides
- Types
 - **Hydration**: the force of running water wearing down rocks.
 - **Corrosion**: chemical action that leads to weathering.
 - **Attrition**: river load particles colliding against each other
 - **Corrasion or abrasion**: solid river load striking against rocks and wearing them down.



Fluvial processes.....EROSION

- Fluvial erosion **proceeds** in two ways:
- vertical erosion: a river erodes its river bed, i.e. it is deepen.
 - lateral erosion: a river erodes its bank; i.e. the river broadens.



 internet geography



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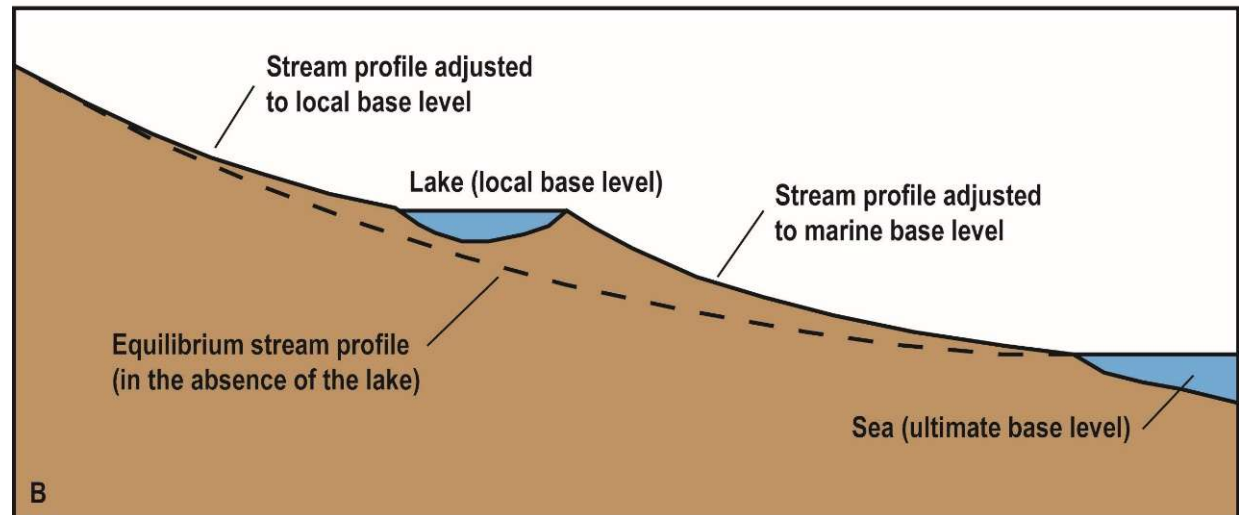
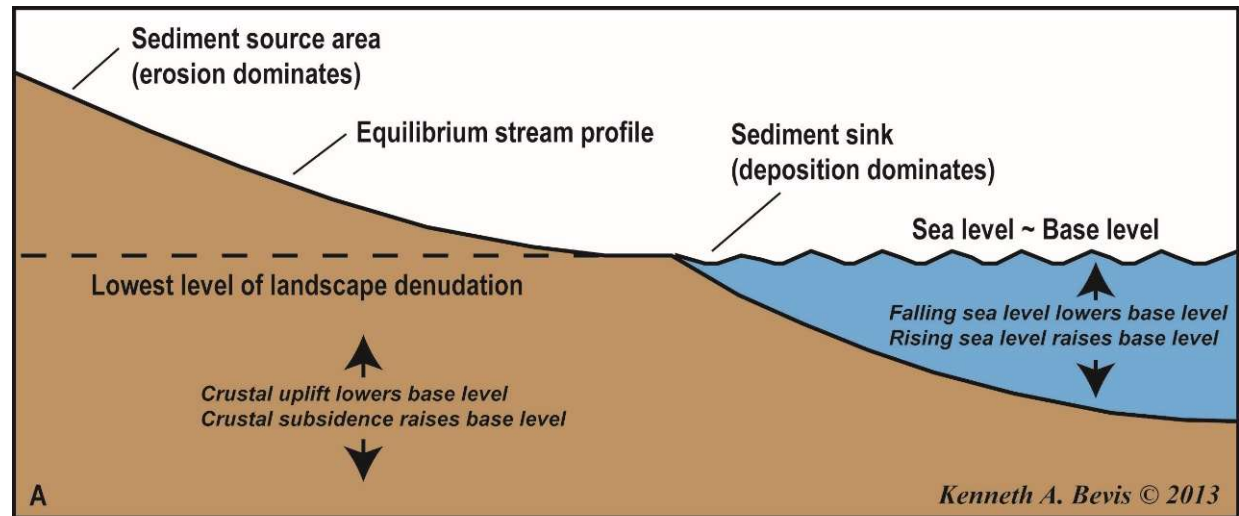
- **Depends** on
 - Flow energy = $f(\text{water depth, slope})$
 - Grain size
 - Base level of erosion



- **Base level of erosion**

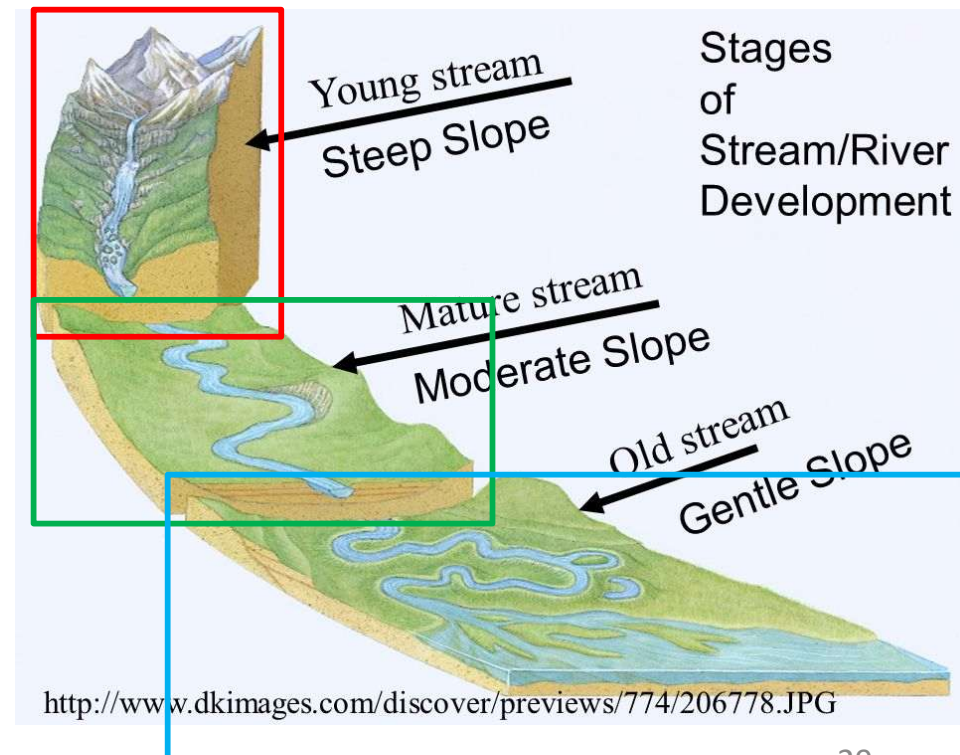
- Lower limit for an erosion process.
- Base level change may be related to the following factors:

- Sea level change
- Tectonic movement
- Extensive sedimentation



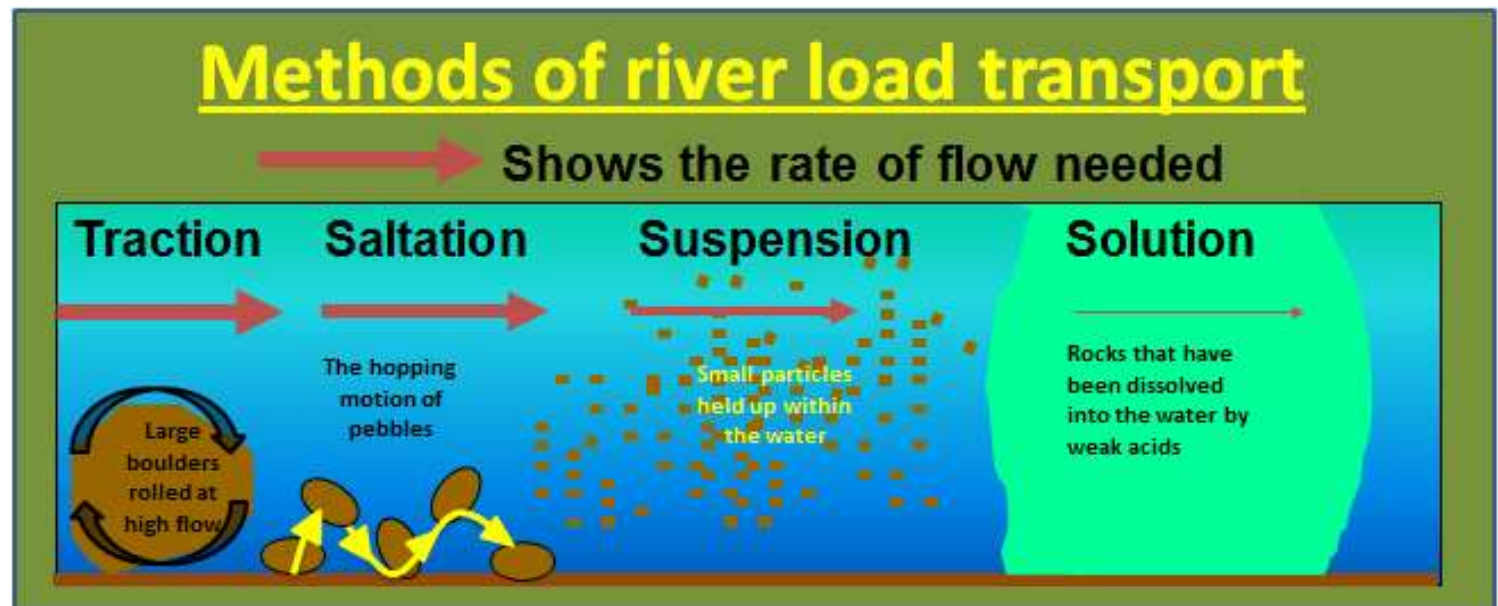
Cycle of erosion.....

Stage	Landscape
Youthful	• steep hillsides,
	• V-shaped valleys
Mature	• rounded hills,
	• valley walls predominant
	• graded streams
	• broad floodplains
Old age	• close to base level
	• very low relief,



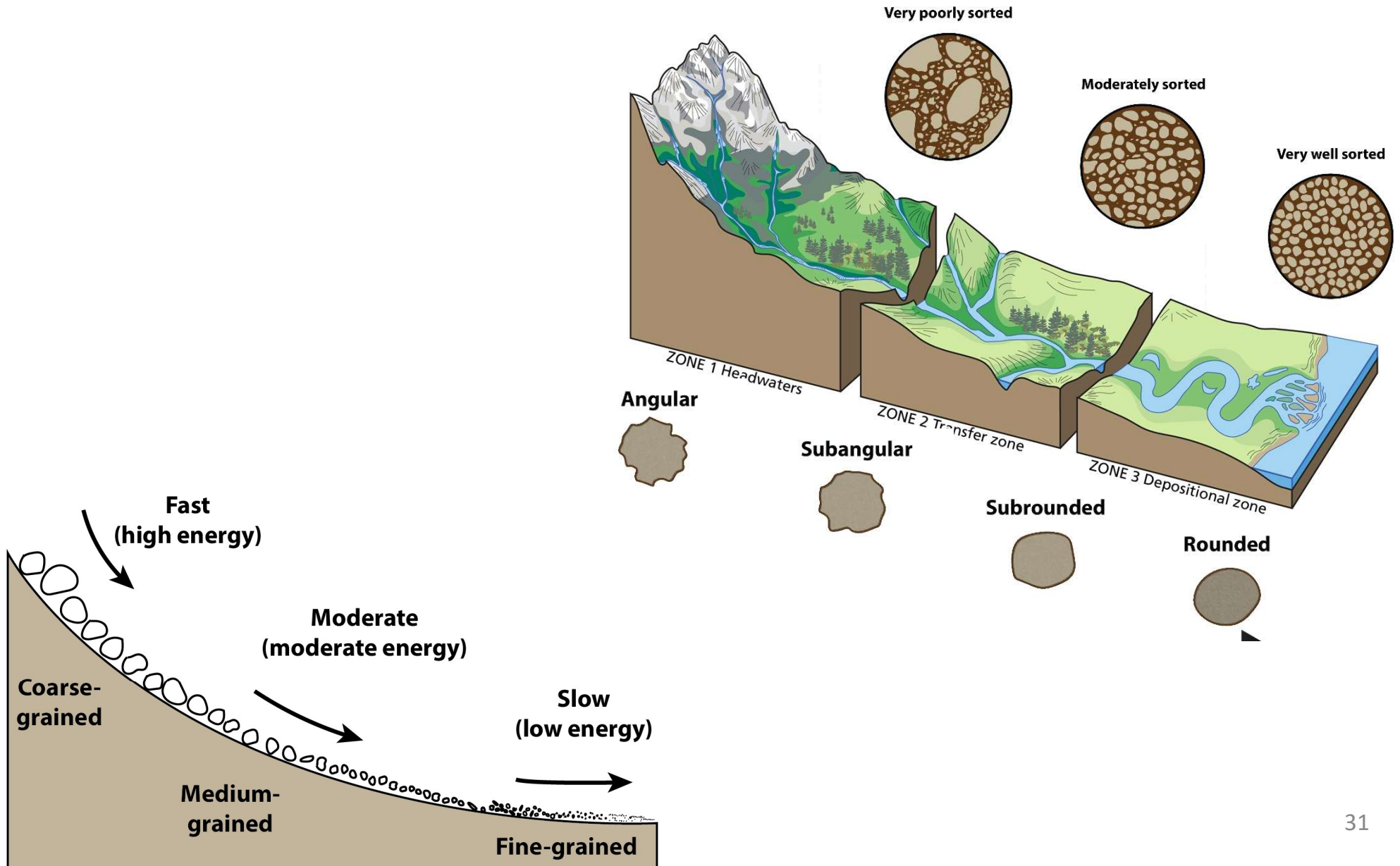
Fluvial processes.....TRANSPORTATION

- **Transportation:** Movement of earth material by river
- 4 ways of transportation
 - **Suspension:** fine light material is carried along in the water
 - **Saltation:** small pebbles and stones are bounced along the river bed,
 - **Traction:** large boulders and rocks are rolled along the river bed,
 - **Solution:** LSt.



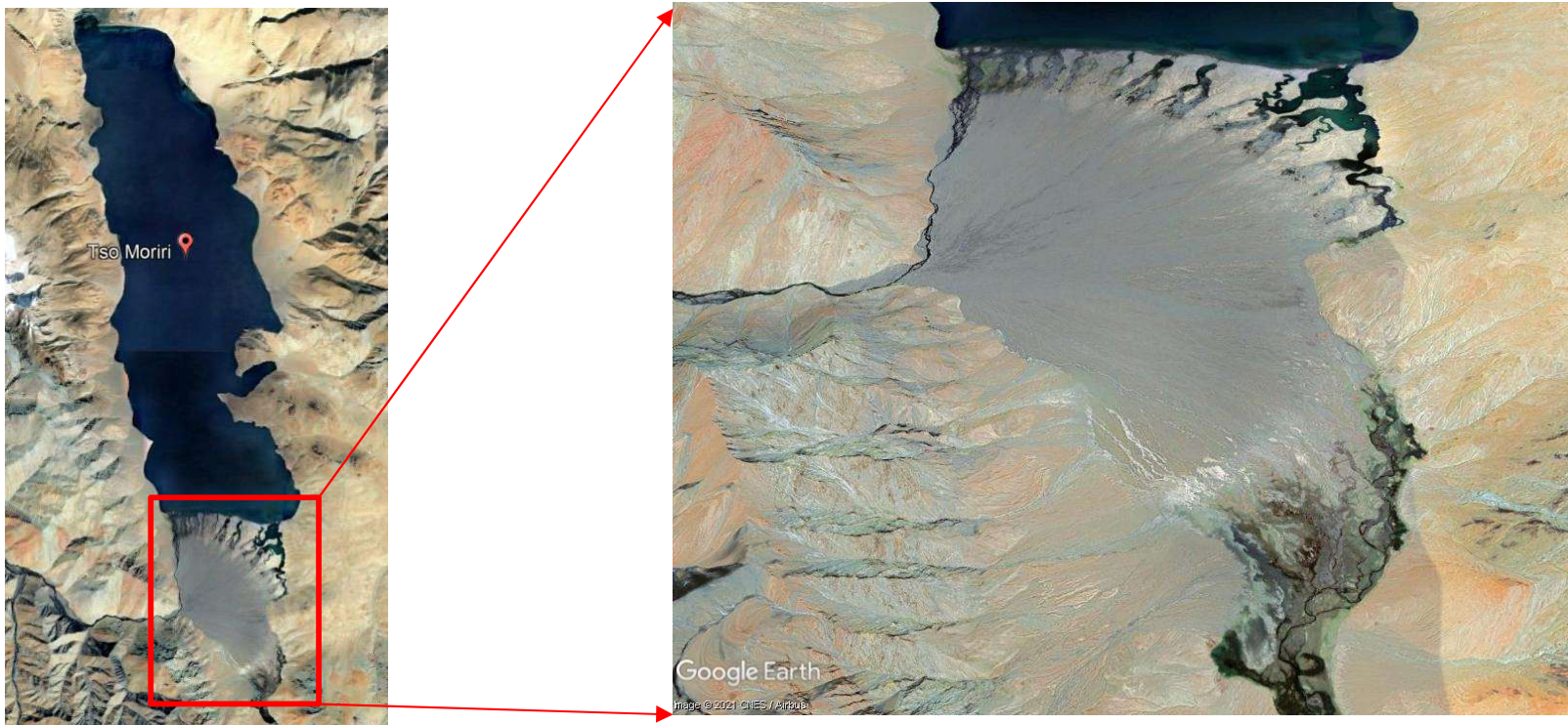
Fluvial processes.....TRANSPORTATION

During the transport the particles are gradually reduced in size and rounded.



Alluvial fan

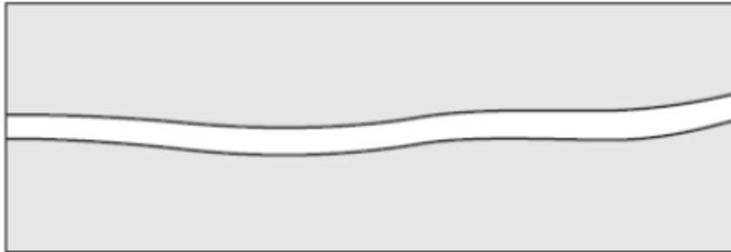
- Fan-shaped deposits
- Alluvial fans are fan-shaped landforms
- Sudden change in landscape, from a steep gradient (such as a mountainous region) to a flat plain.



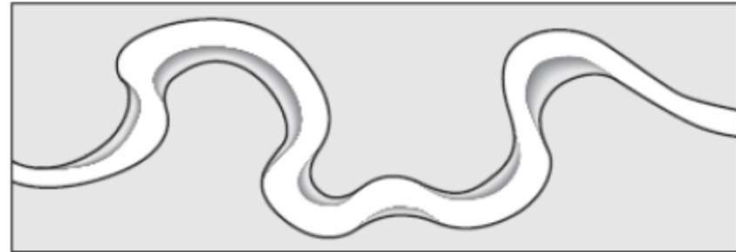
River forms

- Extent of erosion, transportation, and deposition
- Grain size, stream energy, slope, geology....

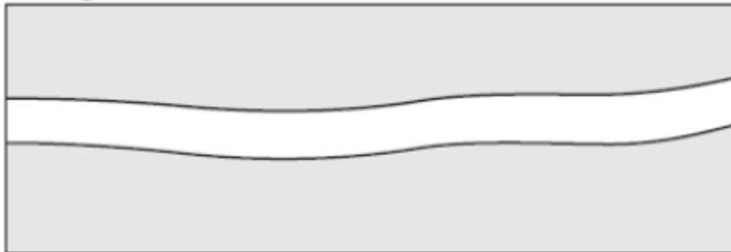
Straight



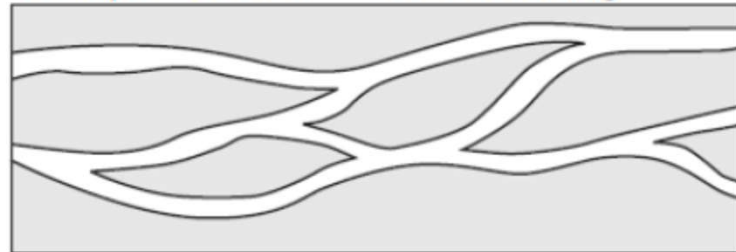
Sinuuous - 'meandering'



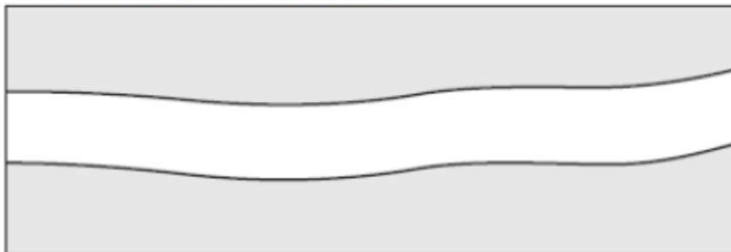
Single channel



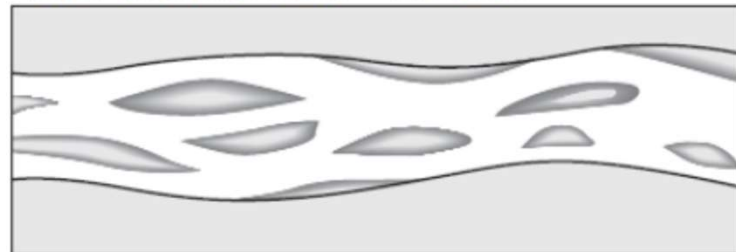
Multiple channel - 'anastomosing'



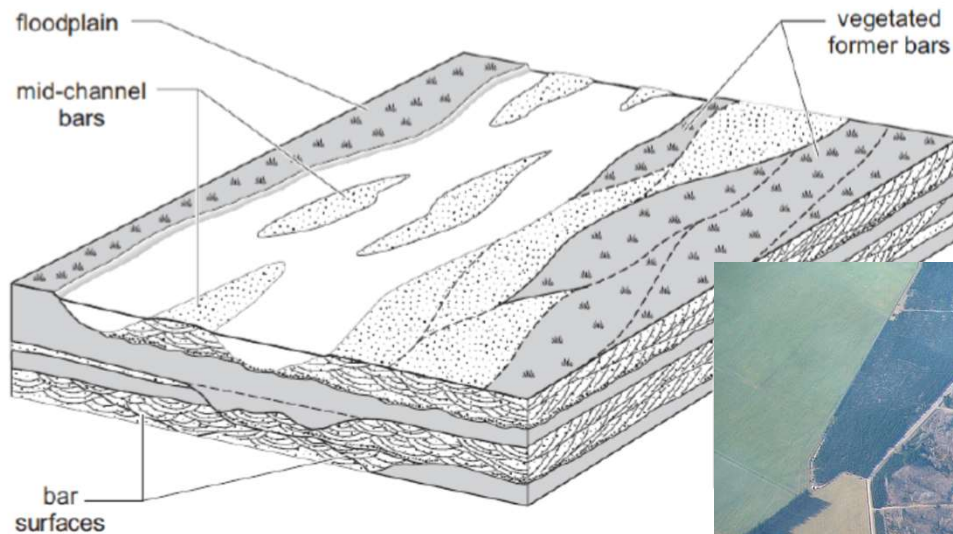
Without channel bars



With channel bars - 'braided'



Network of river channels separated by small, often temporary, islands called braid bars



- Rivers flowing over gently sloping ground begin to curve back and forth across the landscape.
- Channels are asymmetrical.
- The deepest part of the channel is on the outside of each bend.
- The water flows faster : deeper section → erosion water flows slowly : shallow regions → deposition (point bars)

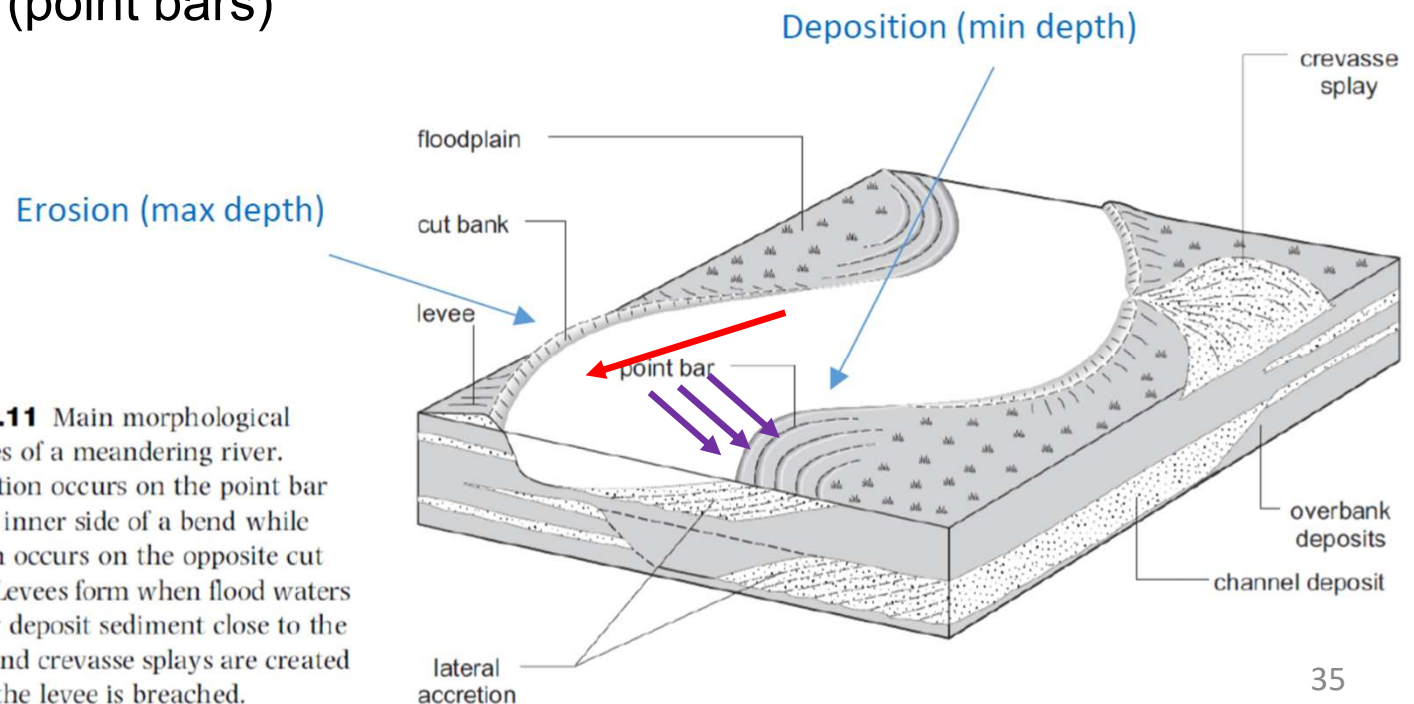
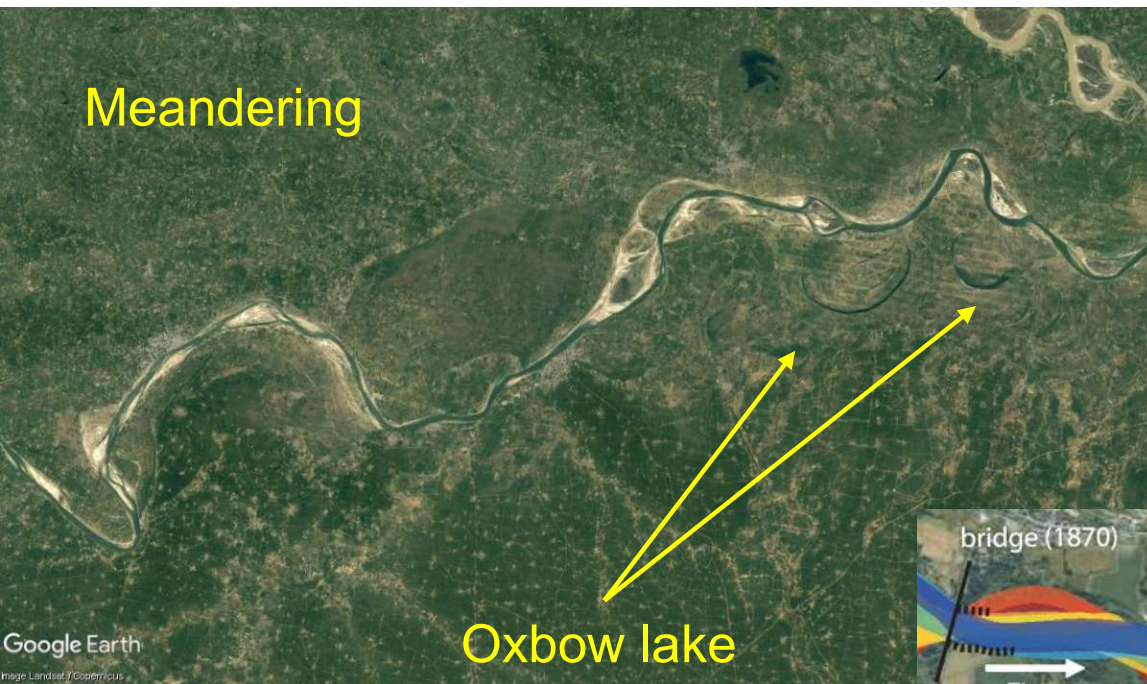
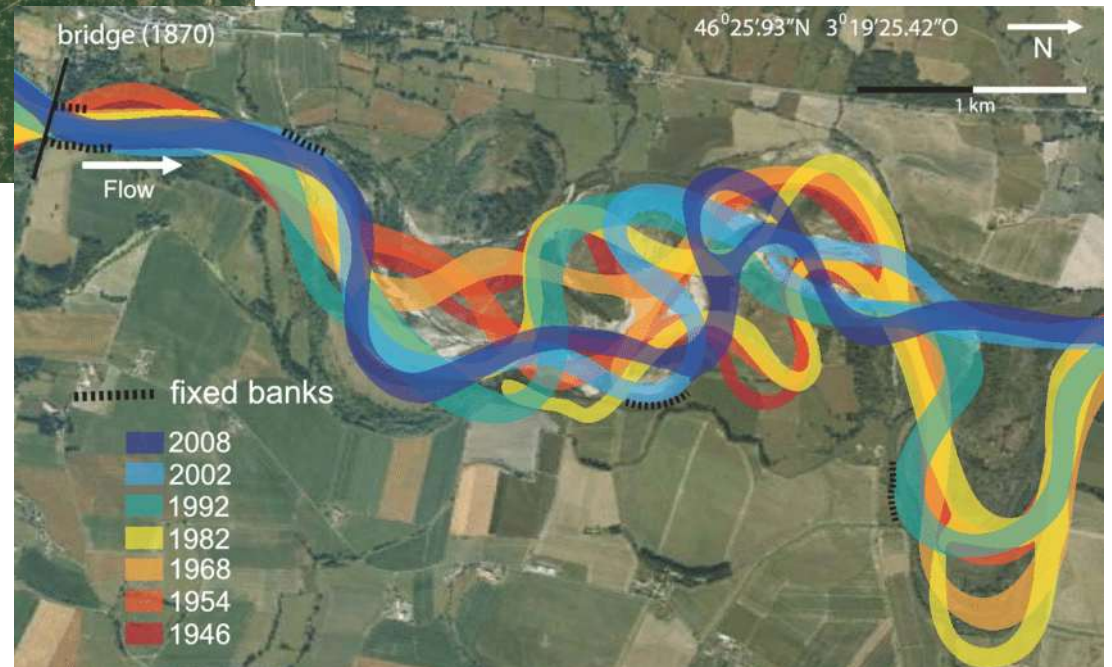


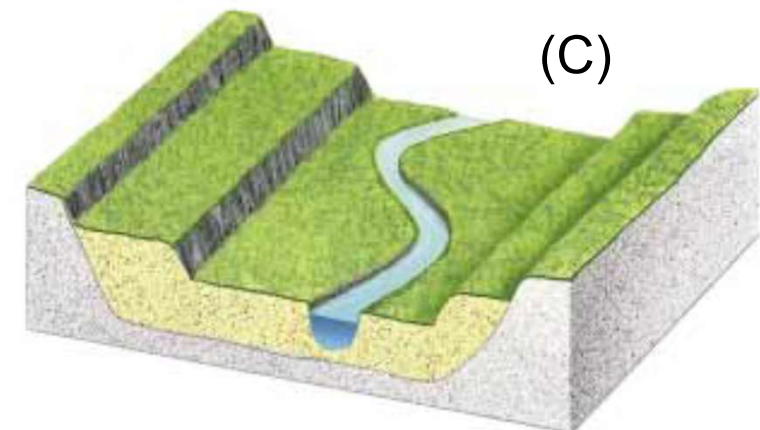
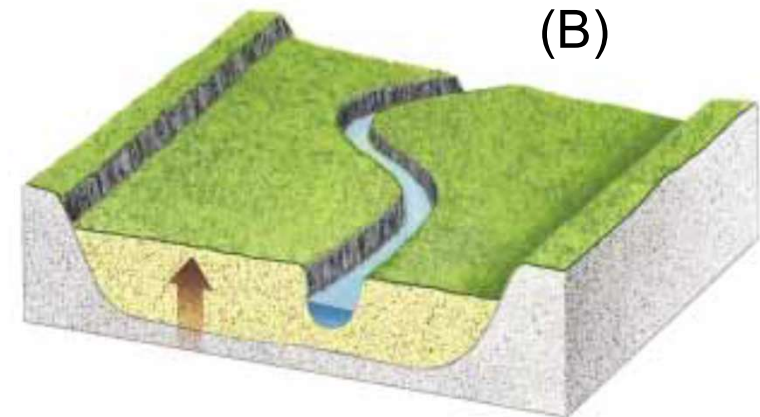
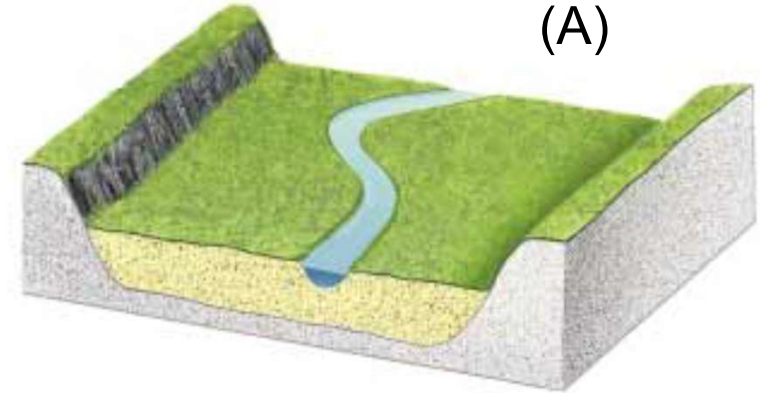
Fig. 9.11 Main morphological features of a meandering river. Deposition occurs on the point bar on the inner side of a bend while erosion occurs on the opposite cut bank. Levees form when flood waters rapidly deposit sediment close to the bank and crevasse splays are created when the levee is breached.



Meandering pattern in
River Allier, France



- Remnants of former floodplain
- River adjusts to uplift or drop in ultimate base level (sea-level) by downcutting
- Older, higher floodplain now a “terrace”
- Steps:
 - River meanders across floodplain
 - Base level drops, or region uplifts.
 - Water goes faster
 - Old, unused floodplain a terrace
 - Further incision cuts a new terrace





Things to remember

- **Fluvial systems: Introduction**

Factors
Drainage basin
Flow velocity
Grain size
Sorting

- **Fluvial processes**

Suspension
Saltation
Traction
Solution

- *Erosion*
- *Transportation*
- *Deposition*

Hydration
Corrosion
Attrition
Corrasion
or abrasion

- **River forms**

- *Braided river*
- *Meandering river*
- *Flood plains*
- *Terraces*

